SIEMENS MICROMASTER VECTOR

Complete Recorded Parameter Set - VFD 7 / Second Outfeed Conveyor

Field values transcribed from the operator's direct keypad readings. Values are recorded settings or live read-only snapshots, not assumed factory defaults.

P Number	Value	Siemens function / interpretation
P001	**0**	Display mode - output frequency
P002	**1.5**	Ramp-up time (s)
P003	**1**	Ramp-down time (s)
P004	**0**	Smoothing time (s)
P005	**5**	Digital frequency setpoint (Hz)
P006	**2**	Frequency source - fixed frequencies
P007	**0**	Keypad control disabled / terminal control
P009	**3**	All parameters readable and adjustable
P010	**1**	Display scaling
P011	**0**	Frequency setpoint memory disabled
P012	**0**	Minimum motor frequency (Hz)
P013	**60**	Maximum motor frequency (Hz)
P014	**0**	Skip frequency 1 (Hz)
P015	**0**	Automatic restart after mains failure disabled
P016	**0**	Start on the fly disabled
P017	**1**	Continuous smoothing
P018	**0**	Automatic restart after fault disabled
P019	**2**	Skip-frequency bandwidth (Hz)
P021	**0**	Minimum analogue frequency (Hz)
P022	**60**	Maximum analogue frequency (Hz)
P023	**0**	Analogue input 1 type - 0-10 V / 0-20 mA
P024	**0**	No analogue setpoint addition
P025	**0**	Analogue output 1 - output frequency
P027	**0**	Skip frequency 2 (Hz)
P028	**0**	Skip frequency 3 (Hz)
P029	**0**	Skip frequency 4 (Hz)
P031	**60**	Jog frequency right (Hz)
P032	**5**	Jog frequency left (Hz)
P033	**1**	Jog ramp-up time (s)
P034	**1**	Jog ramp-down time (s)
P040	**0**	Positioning function disabled
P041	**0**	Fixed frequency 1 (Hz)
P042	**0**	Fixed frequency 2 (Hz)
P043	**0**	Fixed frequency 3 (Hz)
P044	**56**	Fixed frequency 4 (Hz)
P045	**0**	Fixed-frequency 1-4 inversion selection
P046	**55**	Fixed frequency 5 (Hz)

P Number	Value	Siemens function / interpretation
P047	**0**	Fixed frequency 6 (Hz)
P048	**0**	Fixed frequency 7 (Hz)
P049	**0**	Fixed frequency 8 (Hz)
P050	**1**	Fixed-frequency 5 inverted; 6-8 normal
P051	**18**	DIN1 terminal 5 - fixed frequency 5 plus RUN
P052	**18**	DIN2 terminal 6 - fixed frequency 4 plus RUN
P053	**0**	DIN3 terminal 7 disabled
P054	**0**	DIN4 terminal 8 disabled
P055	**0**	DIN5 terminal 16 disabled
P056	**0**	Digital-input debounce - 12.5 ms
P057	**1**	Digital-input watchdog interval (s)
P061	**6**	Relay RL1 - fault indication
P062	**8**	Relay RL2 - warning active
P063	**1**	External-brake release delay (s)
P064	**1**	External-brake stopping time (s)
P065	**1**	Current threshold for relay (A)
P066	**100**	Compound braking (%)
P069	**1**	Ramp extension enabled
P070	**4**	Braking-resistor duty cycle - 100% (MMV only)
P071	**0**	Slip compensation (%)
P072	**250**	Slip limit (%)
P073	**0**	DC injection braking (%)
P074	**1**	I2t motor protection curve
P075	**0**	External braking resistor not connected
P076	**0**	Pulse frequency - 16 kHz, normal modulation
P077	**1**	FCC control mode
P078	**100**	Continuous boost (%)
P079	**50**	Starting boost (%)
P080	**0.80**	Motor power factor
P081	**60**	Motor nameplate frequency (Hz)
P082	**1725**	Motor nameplate speed (RPM)
P083	**3.2**	Motor nameplate current (A)
P084	**230**	Motor nameplate voltage (V)
P085	**1**	Motor nameplate power (hp with P101=1)
P086	**150**	Motor current limit (%)
P087	**0**	Motor PTC disabled
P088	**0**	Automatic stator calibration inactive
P089	**4.6**	Stator resistance (ohm)
P091	**0**	USS serial slave address
P092	**6**	USS baud rate - 9600
P093	**0**	Serial timeout disabled
P094	**60**	USS nominal system setpoint (Hz)

P Number	Value	Siemens function / interpretation
P095	**0**	USS compatibility - 0.1 Hz resolution
P099	**0**	No option module
P101	**1**	North American operation - 60 Hz / hp
P111	**1.01**	Inverter power rating (hp), read-only
P112	**6**	Inverter type - MICROMASTER Vector
P113	**14**	Drive model - MMV75/2
P121	**1**	RUN button enabled when P007=1
P122	**1**	Forward/reverse button enabled when P007=1
P123	**1**	JOG button enabled when P007=1
P124	**1**	Up/down buttons enabled when P007=1
P125	**1**	Forward and reverse operation allowed
P128	**120**	Fan switch-off delay (s)
P131	**0**	Frequency setpoint (Hz), read-only snapshot
P132	**0**	Motor current (A), read-only snapshot
P133	**0**	Motor torque (%), read-only snapshot
P134	**303**	DC-link voltage (V), read-only snapshot
P135	**0**	Motor speed (RPM), read-only snapshot
P137	**0**	Output voltage (V), read-only snapshot
P138	**0**	Rotor/shaft frequency (Hz), read-only snapshot
P139	**0**	Peak output-current record
P140	**231**	Most recent fault - F231
P141	**231**	Previous fault - F231
P142	**231**	Second previous fault - F231
P143	**231**	Third previous fault - F231
P186	**200**	Instantaneous motor-current limit (%)
P201	**0**	PID closed-loop mode disabled
P202	**1**	PID proportional gain
P203	**0**	PID integral gain
P204	**0**	PID derivative gain
P205	**1**	PID sample interval - 25 ms
P206	**0**	PID transducer filtering off
P207	**100**	PID integral capture range (%)
P208	**0**	Transducer type - direct acting
P210	**1.4**	Transducer reading (%), read-only snapshot
P211	**0**	PID 0% setpoint
P212	**100**	PID 100% setpoint
P220	**0**	Frequency cutoff disabled
P321	**0**	Analogue-input-2 minimum frequency (Hz)
P322	**60**	Analogue-input-2 maximum frequency (Hz)
P323	**0**	Analogue input 2 - 0-10 V / 0-20 mA
P356	**6**	DIN6 terminal 17 - fixed frequency 6
P386	**1**	Vector speed-loop proportional gain

P Number	Value	Siemens function / interpretation
P387	**1**	Vector speed-loop integral gain
P720	**0**	Direct I/O functions - normal operation
P721	**0**	Analogue-input-1 voltage (V), read-only
P722	**0**	Analogue-output-1 current (mA)
P723	**0**	Digital-input state (hex) - all inputs off
P724	**0**	Direct relay control - both off
P725	**0**	Analogue-input-2 voltage (V), read-only
P910	**0**	Local control mode
P922	**2.04**	Software version, read-only
P923	**0**	Equipment system/reference number
P930	**231**	Most recent fault - F231 (copy of P140)
P931	**0**	Most recent warning - none
P944	**0**	Factory reset inactive
P971	**1**	EEPROM storage enabled

Critical retained readings: P082 = 1725 RPM; P128 = 120 s; P922 = software 2.04; P140-P143 and P930 = F231. P944 must remain 0 unless an intentional factory reset is required.

Literature: Siemens MICROMASTER Vector / MIDIMASTER Vector Operating Instructions, G85139-H1751-U529-D1; compact guide G85139-H1751-U568-D1, May 1999.

VFD 7 WIRING CROSS-REFERENCE

Second Outfeed Conveyor - confirmed field wire landings

Wire	Siemens landing	Function from wiring + parameter set	Allen-Bradley cross-reference	Field notes
6	**Control terminal 2**	Control 0 V / signal common	Rack ___ Slot ___ Point ___	Source/common: ___________________
128	**Control terminal 5 - DIN1**	P051=18: RUN + fixed frequency 5; P046=55 Hz; P050=1 reverses FF5	Rack ___ Slot ___ Point ___	Relay/terminal: __________________
129	**Control terminal 6 - DIN2**	P052=18: RUN + fixed frequency 4; P044=56 Hz; normal direction	Rack ___ Slot ___ Point ___	Relay/terminal: __________________
188	**Control terminal 19 - RL1**	RL1 switched contact; P061=6 fault indication (active-low function)	Rack ___ Slot ___ Point ___	Cross-ref with wire 187
187	**Control terminal 20 - RL1**	RL1 common/contact circuit; P061=6 fault indication	Rack ___ Slot ___ Point ___	Cross-ref with wire 188

POWER AND MOTOR CONDUCTORS

Wire	Drive terminal	Circuit	Upstream/downstream cross-reference
591	**A / L1**	Incoming three-phase line	Panel __________ Disconnect __________ Fuse __________
593	**B / L2**	Incoming three-phase line	Panel __________ Disconnect __________ Fuse __________
595	**C / L3**	Incoming three-phase line	Panel __________ Disconnect __________ Fuse __________
592	**U**	VFD output to motor	Motor/J-box terminal __________ Splice __________
594	**V**	VFD output to motor	Motor/J-box terminal __________ Splice __________
596	**W**	VFD output to motor	Motor/J-box terminal __________ Splice __________
Not recorded	**PE / ground**	Protective earth	Wire ID __________ Ground point __________

Safety/verification note: Confirm conductor IDs and terminal markings de-energized under LOTO before moving the replacement drive wiring. Do not megger conductors while connected to the VFD. The RL1 contact arrangement should be verified against the terminal symbol on this exact drive before energization.

ALLEN-BRADLEY PLC CROSS-REFERENCE WORKSHEET

Use this page while tracing the VFD conductors through interposing relays and the SLC 5/03 I/O rack.

VFD wire	From / to	Relay ID	Rack	Slot	I/O point	PLC address	Drawing rung/page	Verified
6	T2 - 0 V common							[ ]
128	T5 - DIN1 / 55 Hz reverse							[ ]
129	T6 - DIN2 / 56 Hz forward							[ ]
188	T19 - RL1 contact							[ ]
187	T20 - RL1 contact/common							[ ]
								[ ]
								[ ]
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PLC cabinet / rack identification: ____ Drawing set: ____________________________

Verified by: _____________________________________ Date: ___________________ Machine state during verification: ____________________________